DONUT DUTY

Ssuhaan Jaiin / K25057493 / ssuhaan.jaiin@kcl.ac.uk / PPA Coursework : 1

THE GAME

Donut Duty is a donut themed text-based adventure game built upon the Zuul framework. The player of the game plays as ***"Detective Crumb"*** - the protagonist of the game. They have to navigate the map, defeat the thief - ***"Rico Sugarhands"***, and return the three stolen ***Golden Donuts*** to ***Old Man Krusty*** to save his bakery.

The game expands the original Zuul structure of the game by adding

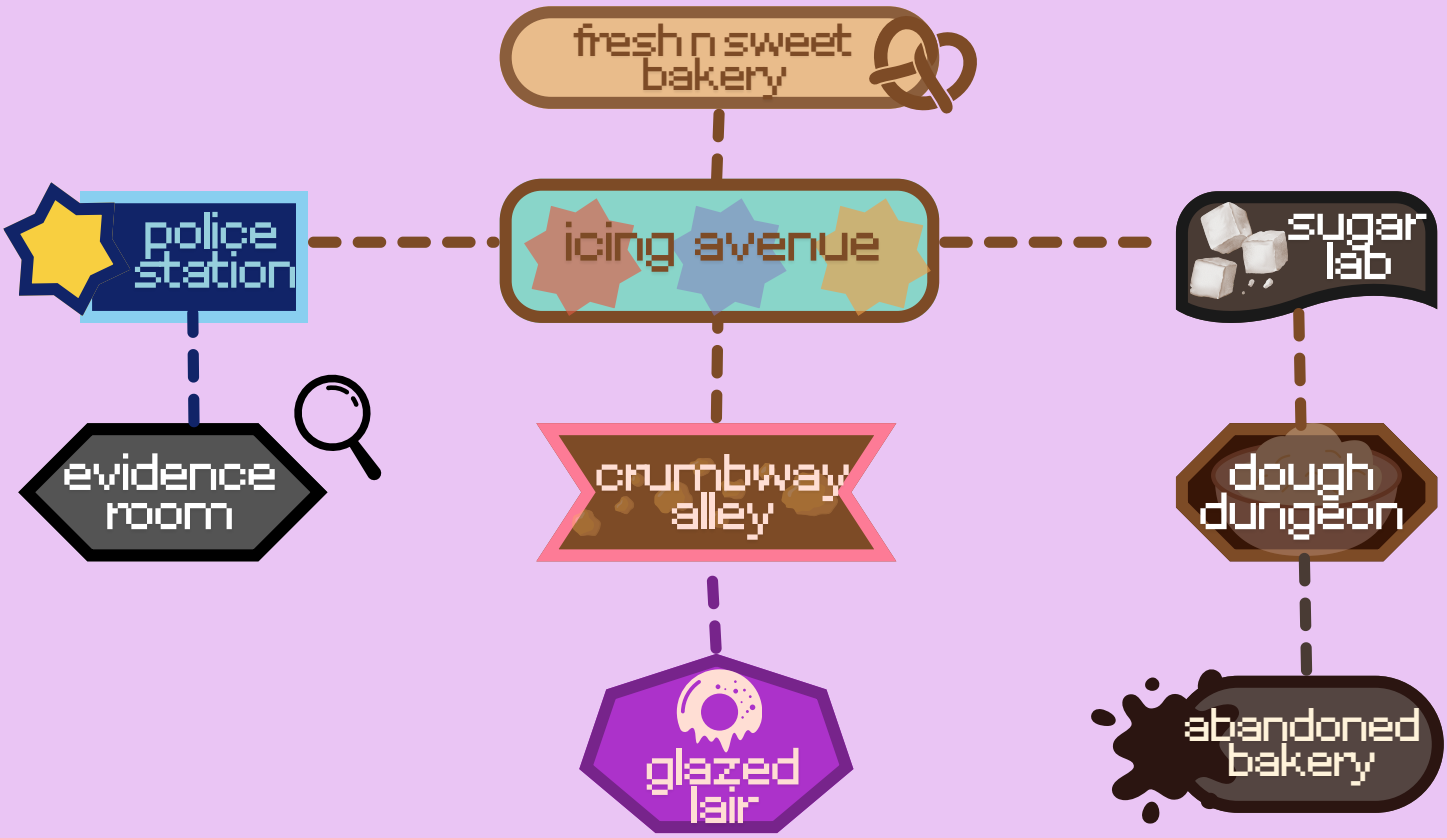
- Multiple interactable NPCs
- Combat mechanics
- Consumables & weapons
- Locked rooms & keys
- Moving NPC Behaviour
- An inventory with a weight limit
- A "back" command
- A teleportation system

USER LEVEL DESCRIPTION

To win the game, the player must

1. Find and collect the ***Golden Donuts*** (located in rooms and held by Rico Sugarhands)
2. Give a ***Reactor Cube*** to ***Dr. Sprinkleford*** to obtain the ***Special Key***
3. Use the ***Special Key*** to access the ***Glazed Lair***
4. Deliver the Golden Donuts to ***Old Man Krusty***

THE MAP



Detective Crumb

You play as **Detective Crumb!** Your task is to explore the map, manage your inventory and health, look for the lost Golden Donuts and help save Old Man Krusty's Abandoned Bakery

Old Man Krusty

Owner of the once famed "Krusty Bakery". Help **Old Man Krusty** save his bakery by giving him the three stolen Golden Donuts. Find him at the Abandoned Bakery

Dr. Sprinkleford

The infamous scientist at the Sugar Lab!
He has misplaced his **Reactor Cube.** If you can help him find it, he might assist you in opening some doorways...

Sgt Cookie

A strict, heavily-armed police officer who patrols Donut City with zero patience. *He attacks anyone* who gets in his way and carries a dangerously oversized and **Overpowered Bazooka.**

Rico Sugarhands

A notorious donut thief and the game's main antagonist. Cunning and dangerous, he hoards a Golden Donut and fights fiercely in the **Glazed Lair.**

KEY CLASSES IMPLEMENTED

- ***Game*** Handles world creation, NPC management, and game flow.
- ***Room*** Stores exits, items, locked state, and NPC presence.
- ***Player*** Manages inventory, health, movement history, and interactions.
- ***NPC*** Supports hostility, movement, combat, inventory, dialogue.
- ***Item*** Supports weapons, consumables, and items.
- ***Command*** Helps break down input text into three key words
- ***Parser*** Helps to take and register input
- ***CommandWords:*** Handles three-word commands and parsing.

BASE TASKS COMPLETED

At least Six rooms

- Fresh & Sweet Bakery
- Police Station
- Icing Avenue (starting room)
- Sugar Lab
- Evidence Room
- Crumbway Alley
- Dough Dungeon
- Glazed Lair (final boss room)
- Abandoned Bakery (Old Man Krusty's location)
- Teleport Room (portal chamber)
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Items in Rooms

Each room may hold multiple items (weapons, consumables)

Carrying System & Weight Limit

The player inventory has a limit of 30 weight units. Items check weight before being picked up.

Win Condition

Winning requires delivering three Golden Donuts to Old Man Krusty after defeating Rico Sugarhands.

The "Back" Command

The Player stores past rooms in an ArrayList, allowing continuous backtracking.

Four Additional Commands

- talk (NPC dialogue)
- eat (consumables)
- attack (combat system)
- give (quest item exchange)

CHALLENGE TASKS IMPLEMENETED

Moving NPCs (Patrol Behaviour)
Sgt Cookie moves between rooms randomly on a patrol route.
If he enters the player's room, he attacks automatically.

Three-word command system
The parser accepts commands such as:
give reactorCube drSprinkleford
attack ricoSugarhands bazooka

Teleportation Room
A dedicated room sends the player to a random unlocked room whenever entered.

ADDITIONAL FEATURES ADDED

Combat System
Weapons with different Combat Power
Hostile auto-attacking NPC
Player and NPC Health
NPC Inventory drop on death

Consumables and Health
Items like donuts and croissants restore health

Locked Rooms and Keys
Allows rooms to be locked and be accessed using a key

Automatic Movement Damage
Players take 5 damage per move

NPC Behavioural States
NPCs can be friendly or hostile which can also change

GAMEPLAY MECHANICS

go ‹ direction ›	*go to the list of directions available, shows error if unavailable*
back	*go to the previously visited room*
look	*return details of the current room ie exits, items and NPCs*
status	*returns the health and inventory weight to the player*
inventory	*returns the list of items and their weight + descriptions*
take ‹ item ›	*add the item to the inventory*
drop ‹ item ›	*remove the item from your inventory*
eat ‹ item ›	*consume the item if it is consumable*
give ‹ item › ‹ npc ›	*give an item to an NPC*
talk ‹ npc ›	*talk to an NPC*
attack ‹ npc › ‹ weapon ›	*attack an NPC with a weapon*
help	*show available commands*
quit	*quit the game*

Moving to another room makes the player lose 5 health to make the game difficult to play
If you are caught by Sgt, Cookie, then you are dealt 20 damage

CODE QUALITY CONSIDERATIONS

Cohesion

Each class has a single and clear responsibility.
Eg : The player class handles only health and inventory related data and the NPC class handles all NPC related information.

Low Coupling

Low coupling is achieved by reducing the dependencies of varios classes on each other.
Eg : NPC combat does not modify the Game class directly

Responsibility Driven Design

All objects are able to handle their logic via their own methods
Eg : NPC.attack() is used to handle logic related to damaging

Maintainability

The code has been written in a way to support expansion. It allows the easy addition of NPCs, rooms, items, behaviours and conditions

FINAL WALKTHROUGH

Some aspects of the game such as interacting with an NPC is based on luck and chance, hence there is no shortest playthrough.
Each move made by the player causes a deduction in their health by 5 HP to make the game interesting and challenging, hence eating and attacking is considered essential

A potential playthrough can look like this :
take donut, go north, take sugarDonut, take chocolateDonut, take croissant, go south, go west, take gun, go south, eat chocolateDonut, eat donut, take goldenDonut, take reactorCube, take donut, take sugarDonut, go north, go east, go east, give reactorCube drSprinkleford, go west, go south, take goldenDonut, go south, eat croissant, eat donut, eat sugarDonut, attack ricoSugarhands gun, attack ricoSugarhands gun, take donut, eat donut, attack ricoSugarhands gun, take goldenDonut, go north, go north, go east, go south, go south, give goldenDonut oldManKrusty, give goldenDonut oldManKrusty, give goldenDonut oldManKrusty

CONCLUSION

Programming as a task seems daunting at first and it takes a good amount of practice to develop code that is both well-implemented and effective. Completing “Donut Duty” as my first coursework has given me the confidence to successfully take on the challenges of programming and problem solving and excites me for the content yet to be covered in PPA and Computer Science.

(No generative LLM used to produce source code)
(Referred to PPA material on KEATS, w3schools to check for ArrayList methods for Room class and NPC movement)